

Enabling Augmented Reality Incorporate with Audio on Indoor Navigation for People with Low Vision

Zihao Chi*, Zhaofeng Niu†, Taishi Sawabe‡
Interactive Media Design Laboratory
Nara Institute of Science and Technology

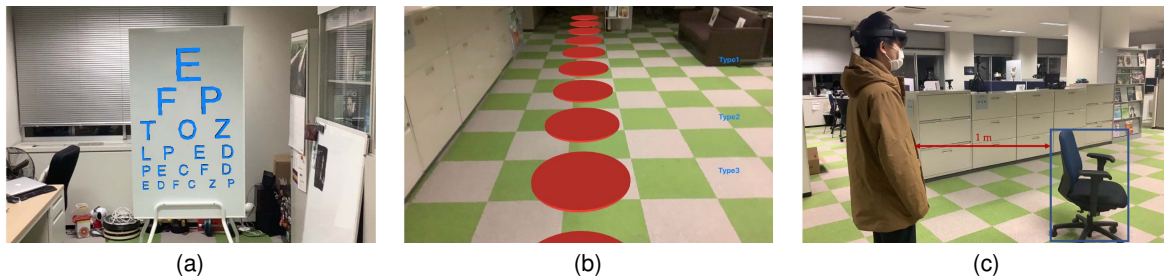


Figure 1: Three main parts of the indoor navigation application: (a) Visual rating measurement by Snellen chart; (b) Virtual guidance for navigation; (c) Object detection and depth estimation result.

ABSTRACT

Effective indoor navigation is a challenging task for people with low vision, even they can potentially benefit from visual cue. Many researchers have proposed various approaches to improve the efficiency and accuracy of the navigation, for example, using audio cue as an assist. However, the visual rating has been neglected, which is important to provide high-quality assistant for different visual impairments. In this paper, we propose an augmented reality (AR) application that measures the visual rating. According to the visual rating, different navigation services, including visual aid and audio cue, are provided to users. Besides, object detection and depth estimation are adopted for detecting and avoiding obstacles, which can significantly improve the performance of the navigation. For the experiment, the Snellen chart is displayed on Microsoft HoloLens2 and a pilot study has been conducted to verify its feasibility. The guidance information are shown in various sizes, combining with audio cue, for different vision-rating users. The strategy on both visual aid and audio cue will be tested by a user study for the next step. The results of object detection and depth estimation have shown high accuracy on Microsoft HoloLens2 and will be integrated into the application in the future.

Index Terms: Low vision; Visual rating; Augmented reality; Navigation application

1 INTRODUCTION

There is no doubt that the eye is one of the most important components of human body and vision plays an essential role in understanding the environment. However, there are many people with low vision. According to the report by WHO [2], at least 2.2 billion people have a near or distance vision impairment in the world, and almost half of these cases, low vision problems could have been prevented or have yet to be addressed while they may meet a lot of

difficulties in their daily lives. An effective navigation application, which provides accurate navigation information and help to avoid obstacles, is a great tool for people who have vision troubles.

Many researchers have proposed and developed various navigation applications, most of which are based on audio instruction to guide the people with visual impairment. For example, Microsoft Soundscape [1] enables the users to build a richer understanding of their surroundings. Ahmetovic et al. proposed NavCog [3], a smartphone-based turn-by-turn navigation application for the blind users. However, these prior researches mainly focus on the blind people, and only audio cue is provided. In fact, people with low vision can also use some visual cues. Zhao et al. [8] proposed visual and audio wayfinding guidance for blind and sighted people. Huang et al. [5] proposed a sign-reading application which could assist visually impaired users. Although some visual aids are proposed, most of the research still fail to consider about the visual rating of users. In this work, we mainly focus on the people with low vision and design the strategy for different vision ratings.

Specifically, we propose an Augmented Reality (AR) application that measures the visual impairment rating before providing the visual solutions. In the application, we adopt the Snellen chart (as shown in Fig.1 (a)), which is one of the common-used tools for quantify the visual acuity, to perform the low vision rating. Different from the traditional way, the chart is displayed virtually on AR devices. According to the visual rating, the application provides various types of visual strategies (Fig.1 (b) illustrates one of the strategies) with audio cue. Simultaneously, two computer vision technologies, including object detection and depth estimation, are utilized to detect and avoid the obstacles during the navigation, as shown in Fig.1 (c).

In sum, our idea can be described in three-fold:

- We propose a concept that different visual ratings should be matched to different strategies on visual cues, and the visual rating can be tested on an AR device.
- Aiming to achieve better service for people with low vision, both visual and audio cue are adopted to improve the efficiency and accuracy of the navigation.
- Object detection and depth estimation technologies are utilized to detect obstacles and estimate the distance between the user and the obstacle.

*e-mail: chi.zihao.bv4@is.naist.jp

†e-mail: niu.zhaofeng.mv7@is.naist.jp

‡e-mail: t.sawabe@is.naist.jp

2 OUR IDEA

There are three main parts in our AR application, that is, visual rating measurement, visual cue strategy design and scene understanding.

Visual rating measurement: The Snellen chart is adopted to quantify the visual acuity. We follow the standard instruction but with AR devices. As shown in Fig.1 (a), the block letters are visual objects shown in the scene. The user can be tested along with the audio instruction. Compared with the traditional method, our application has several advantages: 1) It can be used wherever the user is; 2) It is easy to set the specific distance between the Snellen chart and user with AR devices; 3) The audio instruction is clear to understand so that the user can complete the testing independently. According to the definition proposed by WHO [2], three types of visual impairment are considered, *i.e.*, mild, moderate and severe, as shown in Tab.1.

Table 1: Snellen fractions and strategies for ratings of the low vision.

Rating	Snellen Fraction	Virtual Pie Diameter
Mild	<6/12 m	0.1m
Moderate	<6/18 m	0.2m
Severe	<6/60 m	0.4m

Strategy design: For the people with low vision, it is promising to enhance their visual capabilities by augmented reality information [7]. Magnifying the virtual tags is one of an efficient method of compensating for visual loss. Here, we utilize virtual pie as the guidance information and change its size for different visual-rating users. As illustrated in Fig. 2, the diameter of virtual pie becomes larger if the rating is getting worse. The distance between each two virtual pies is fixed. Besides, audio cues, as the additional information, are synchronized released during navigation.



Figure 2: Different virtual pie diameter. From left to right: 0.1m, 0.2m and 0.4m.

Scene understanding: Although we could obtain the map of the indoor environment in advance, there may exist some obstacles on the way. To avoid the obstacles, two of computer vision technologies, *i.e.*, object detection and depth estimation, are adopted for better understanding the scenes. Specifically, object detection technology is used to detect the obstacles and depth estimation technology is adopted to estimate the distance between the user and the obstacles.

3 IMPLEMENTATION AND EXPERIMENT

The application is developed in *Unity* and configured on Microsoft HoloLens2 that provides mixed reality and audio functions. Three main parts of our application has been completed separately: 1) The Snellen Chart can be displayed on HoloLens with audio instruction, which has been tested with 3 myopic participants; 2) different sizes of the virtual pie are provided to show the route incorporate with audio cues; 3) YOLO4 method [4] is adopted to perform object detection and HMA-Depth method [6] is used for depth estimation (Fig.3 shows an example for the results).

In the next step, we will firstly integrate these three parts, then conduct a user study. We plan to recruit participants ($n = 20$), who



Figure 3: Scene understanding results: raw (left), object detection result (middle), depth estimation result (right).

suffers from myopia, hyperopia and presbyopia. In a specific environment where the map is obtained in advance, the participant takes the visual rating measurements following the audio instruction. Next, the participant will be instructed by the virtual pie and audio cue to walk from one point to another. To measure the performance of the navigation, two metrics will be considered, *i.e.*, navigation time and error rate. Besides, we will distribute some questionnaires, collecting the feedback to improve the design and performance of the application.

4 CONCLUSION AND FUTURE WORK

In this paper, we introduce an AR indoor navigation application for people with low vision. It firstly measures the visual rating of users, then provides customized designed visual and audio solutions for each user. Also, two of computer vision technologies, *i.e.*, object detection and depth estimation, are used for detecting and avoiding the obstacles during the navigation. In the future, we will improve the user interface design including the virtual Snellen chart and virtual tags, and integrate all the functions into the application. A user study will be conducted to verify the effectiveness of the application and improve the performance according to the feedback.

ACKNOWLEDGMENTS

This work is supported by Creative and International Competitiveness Project (CICP) of Nara Institute of Science and Technology (NAIST).

REFERENCES

- [1] Microsoft soundscape. Available online. <https://www.microsoft.com/en-us/research/product/soundscape/>.
- [2] Blindness and vision impairment. Available online, October 14, 2021. <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>.
- [3] D. Ahmetovic, C. Gleason, C. Ruan, K. Kitani, H. Takagi, and C. Asakawa. NavCog: A navigational cognitive assistant for the blind. In *Proceedings of the 18th International Conference on Human-Computer Interaction with Mobile Devices and Services*, MobileHCI '16, p. 90–99. Association for Computing Machinery, New York, NY, USA, 2016. doi: 10.1145/2935334.2935361
- [4] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao. Yolov4: Optimal speed and accuracy of object detection, 2020.
- [5] J. Huang, M. Kinader, M. J. Dunn, W. Jarosz, X.-D. Yang, and E. A. Cooper. An augmented reality sign-reading assistant for users with reduced vision. *PloS one*, 14(1):e0210630, 2019.
- [6] Z. Niu, Y. Fujimoto, M. Kanbara, and H. Kato. Hma-depth: A new monocular depth estimation model using hierarchical multi-scale attention. In *2021 17th International Conference on Machine Vision and Applications (MVA)*, 2021. doi: 10.23919/MVA51890.2021.9511345
- [7] L. Stearns, L. Findlater, and J. E. Froehlich. Design of an augmented reality magnification aid for low vision users. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility*, ASSETS '18, p. 28–39. Association for Computing Machinery, New York, NY, USA, 2018. doi: 10.1145/3234695.3236361
- [8] Y. Zhao, E. Kupferstein, H. Rojnirun, L. Findlater, and S. Azenkot. The effectiveness of visual and audio wayfinding guidance on smartglasses for people with low vision. In *Proceedings of the 2020 CHI conference on human factors in computing systems*, pp. 1–14, 2020.